

Problem 73 (Bosnia and Herzegovina TST 2016 Problem 4). Find the largest positive integer n which cannot be expressed as the sum of three pairwise coprime integers larger than 1.

Solution. The answer is 17.

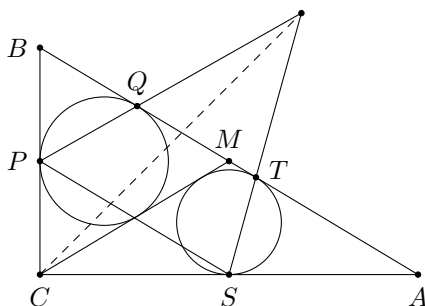
Firstly, if $17 = a + b + c$ where a, b, c are pairwise coprime integers larger than 1, they must be odd since at most one of them can be even. The only possibility is $17 = 3 + 5 + 9$, but then $(3, 9) = 3 > 1$.

Secondly, we show that all integers $n > 17$ have such representations. In the following, k denotes an integer.

- If $n = 4k \geq 20$, then $n = 2 + (2k - 3) + (2k + 1)$, where $2 < 2k - 3 < 2k + 1$, and $2, 2k - 3, 2k + 1$ are pairwise coprime since $2 \nmid 2k - 3, 2k + 1$ and $(2k - 3, 2k + 1) = (2k - 3, 4) = 1$.
- If $n = 4k + 2 \geq 18$, then $n = 2 + (2k - 1) + (2k + 1)$, where $2 < 2k - 1 < 2k + 1$, and $2, 2k - 1, 2k + 1$ are pairwise coprime as above.
- If $n = 12k + 1 \geq 25$, then $n = 3 + (6k - 7) + (6k + 5)$, where $3 < 6k - 7 < 6k + 5$, and $3, 6k - 7, 6k + 5$ are pairwise coprime since $3 \nmid 6k - 7, 6k + 5$ and $(6k - 7, 6k + 5) = (6k - 7, 12) = 1$ (the same arguments apply to the cases below).
- If $n = 12k + 3 \geq 27$, then $n = 3 + (6k - 1) + (6k + 1)$.
- If $n = 12k + 5 \geq 29$, then $n = 3 + (6k - 5) + (6k + 7)$.
- If $n = 12k + 7 \geq 19$, then $n = 3 + (6k - 1) + (6k + 5)$.
- If $n = 12k + 9 \geq 21$, then $n = 3 + (6k + 1) + (6k + 5)$.
- If $n = 12k + 11 \geq 23$, then $n = 3 + (6k + 1) + (6k + 7)$.

The proof is complete. □

Problem 74 (Sharygin Geometry Olympiad 2012 Correspondence Round Problem 8). In $\triangle ABC$, $\angle C = 90^\circ$. Let M be the midpoint of AB . The incircle of $\triangle BCM$ touches BC and BM at P and Q respectively. The incircle of $\triangle ACM$ touches AC and AM at S and T respectively. Prove that PQ , ST and the internal angle bisector of $\angle ACB$ are concurrent.



Solution. It is well-known that $MC = MA = MB$. Since $\triangle MCA$ is isosceles, S is the midpoint of AC . Similarly, P is the midpoint of BC . By the midpoint theorem, we have $SP \parallel AB$. Thus, we have $\angle TSP = \angle STA = \angle TSA$. This implies ST is the external angle bisector of $\angle CSP$. Similarly, PQ is the external angle bisector of $\angle CPS$. Thus, ST and PQ intersect at the C -excentre of $\triangle CSP$, which must lie on the internal angle bisector of $\angle SCP$ as desired. □

Problem 75 (Korea MO Final Round 2018 Problem 5). Do there exist polynomials P, Q with integer coefficients such that $\deg P, \deg Q \geq 2022$ and

$$P(Q(x)) = 3Q(P(x)) + 1 \quad (1)$$

for all real numbers x ?

Solution. Yes. If there are two polynomials X and Y with integer coefficients satisfying (1) (where $(P, Q) = (X, Y)$), then by replacing x by $Y(x)$, we obtain

$$X(Y(Y(x))) = 3Y(X(Y(x))) + 1.$$

This implies $(P, Q) = (X \circ Y, Y)$ satisfies (1), and clearly $X \circ Y$ and Y have integer coefficients. Similarly, replacing x by $X(x)$ in (1), we obtain

$$X(Y(X(x))) = 3Y(X(X(x))) + 1.$$

This implies $(P, Q) = (X, Y \circ X)$ satisfies (1), and clearly X and $Y \circ X$ have integer coefficients.

Now, we start with the pair $(X, Y_0) = (2x^2 + 2x, 3x + 1)$, which satisfies (1) as

$$\begin{aligned} X(Y_0(x)) &= 2(3x + 1)^2 + 2(3x + 1) = 18x^2 + 18x + 4, \\ 3Y_0(X(x)) + 1 &= 3(3(2x^2 + 2x) + 1) + 1 = 18x^2 + 18x + 4 = X(Y_0(x)). \end{aligned}$$

From above, each time we can replace (X, Y_n) by $(X, Y_{n+1}) = (X, Y_n \circ X)$ so that (1) is still satisfied. If $\deg X = a$ and $\deg Y_n = b$, then $\deg(Y_n \circ X) = ab$. Therefore, after doing this for 11 times, we obtain a new pair (X, Y_{11}) satisfying (1) with $\deg Y_{11} = 2^{11} > 2022$. Finally, we can replace this by $(X \circ Y_{11}, Y_{11})$ so that it satisfies (1) and $\deg(X \circ Y_{11}) = 2^{12} > 2022$. So we are done. $\square$

Problem 76 (Korea MO Final Round 2018 Problem 3). There is a tournament of $n \geq 7$ players such that every pair of players may play against each other for an arbitrary number of times (possibly 0 times). It is possible that player A has won over player B in one game, and B has won over A in another game. It is known that for every two players, there is a third player who has won over each of them. Prove that for any positive integer k satisfying $2(2^{2^k} - 1) \geq n$, there exists $m \in \{2, 3, \dots, 2k\}$ and players $A_1, A_2, \dots, A_m$ such that A_{i+1} has won over A_i for $i = 1, 2, \dots, m$, where $A_{m+1} = A_1$.

Solution. Consider a directed graph G whose vertices are the players such that an edge is drawn from A to B if A has won over B . Next, we consider another directed graph G' whose vertices are the players such that an edge is drawn from A to B if there is a path of length at most k in G that goes from A to B . If G' is not simple, then there exist two edges $A \rightarrow B$ and $B \rightarrow A$, which correspond to a circuit of length at most $2k$ in G , and we are done. So we now assume G' is a simple directed graph.

Given any vertices $A_1, A_2, \dots, A_{2^k}$ in G , we can find vertices $B_1, B_2, \dots, B_{2^{k-1}}$ (possibly the same) such that there are edges $B_1 \rightarrow A_1, B_1 \rightarrow A_2, B_2 \rightarrow A_3, B_2 \rightarrow A_4, \dots$ in G by the given condition. Similarly, we can find vertices $C_1, C_2, \dots, C_{2^{k-2}}$ such that there are edges $C_1 \rightarrow B_1, C_1 \rightarrow B_2, C_2 \rightarrow B_3, C_2 \rightarrow B_4, \dots$ in G (if $B_1 = B_2$, etc., we take $C_1 = B_1$, so that $C_1 \rightarrow B_1$ collapses to a vertex). Repeating this k times, we eventually find a single vertex X such that there is a path of length at most k in G that goes from X to each A_i . In other words, there are edges $X \rightarrow A_i$ in G' .

Claim. Let $m \geq 1$. Consider a (nonempty) simple directed graph satisfying the following condition: For any m vertices $A_1, A_2, \dots, A_m$, there exists another vertex B such that each $B \rightarrow A_i$ is an edge. Then this graph contains at least $2^{m+1} - 1$ vertices.

Proof. We use induction on m . When $m = 1$, we pick any vertex A . We know that there exists B such that $B \rightarrow A$ is an edge. Also, there exists C such that $C \rightarrow B$ is an edge. Clearly, $C \neq A$ as the graph is simple. This shows the graph has at least 3 vertices. For the inductive step, consider any vertex X . Consider the subgraph formed by all vertices Y such that there is an edge from Y to X . Then for any $m - 1$ vertices $A_1, A_2, \dots, A_{m-1}$ in this subgraph, there exists B such that each $B \rightarrow A_i$ and $B \rightarrow X$ is an edge by the given condition. Since $B \rightarrow X$ is an edge, B also belongs to this subgraph. Thus, this subgraph contains at least $2^m - 1$ vertices by the inductive hypothesis. This means each vertex has in-degree at least $2^m - 1$, and so the total number of edges is at least $(2^m - 1)|V|$ where $|V|$ is the number of vertices. As the graph is simple, we have

$$(2^m - 1)|V| \leq \frac{|V|(|V| - 1)}{2}.$$

This implies $|V| \geq 2^{m+1} - 1$ as desired. The claim follows by induction. $\square$

In view of the earlier discussion, we can take $m = 2^k$ in the claim. Therefore, we have $n \geq 2^{2^k+1} - 1 > 2(2^{2^k} - 1)$, contradicting the given condition. So we are done. $\square$